

RAHIL SHARMA

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Research Profile

Systems researcher interested in how architectural choices shape what computational systems enable, constrain, and imply. My work is grounded in building working artefacts and using them to study security, efficiency, reliability, explainability, and human judgement. I am particularly drawn to interdisciplinary problems where technical design interacts with organizational practice and societal impact.

Research Interests: Privacy-Preserving Systems · Applied Crypto · Trustworthy AI · Distributed and Data-Intensive Systems · Empirical Systems Research · Sociotechnical Computing

EDUCATION

Master of Science (MSc) Computer Science (Big Data Engineering), Joint Degree

Sept 2024 – Jun 2026

Vrije Universiteit Amsterdam (VU) & University of Amsterdam (UvA), Netherlands

GPA: 8.5/10

Executive PG Program in Software Development

Aug 2022 – Oct 2023

International Institute of Information Technology, India

GPA: 3.3/4

Bachelor of Technology (B. Tech), Electronics and Telecommunication Engineering

Aug 2018 – Jun 2022

Symbiosis International University, India

Percentage: 74%

PUBLICATIONS

Research Interests: Interdisciplinary work at the intersection of applied cryptography, large-scale data systems, and machine learning, with emphasis on protocol-level privacy guarantees, empirical security evaluation, and accountable system design.

- R. Sharma. "OBFI: Implementation, Empirical Analysis, and Crossover Characterization of Competing Protocols." MSc Thesis, VU Amsterdam / UvA, supervised by Dr. M. van Dijk (CWI Amsterdam). Publicly available via Zenodo ([DOI: 10.5281/zenodo.22019435](https://doi.org/10.5281/zenodo.22019435)). Companion library [pyobfi](#) available on PyPI. Code: [GitHub repository](#).
- R. Sharma. "Auditable Financial Fraud Detection: Graph Features and Agentic Investigation." [arXiv:2607.19266, 2026](#).
- R. Sharma. "Garbage Collection and Energy Consumption in Java: A Controlled Study Across Workloads and JDKs." [arXiv:2608.19520, 2025](#).

Full publication list, including earlier undergraduate work, available on [Google Scholar](#).

RESEARCH AND PROFESSIONAL EXPERIENCE

Centrum Wiskunde & Informatica (CWI), Amsterdam, NL

Research Assistant: CSY Group (Security and Machine Learning)

Feb 2025 – Jun 2026

- Implemented cryptographic protocols from research papers into working distributed Python systems.
- Translated theoretical designs into executable code, resolving ambiguities and making key implementation decisions.
- Designed and ran experiments across different configurations to evaluate protocol performance.
- Collaborated with supervisors and research team to refine implementations, validate results, and iterate on experimental design.

Vrije Universiteit, Amsterdam, NL

Senior Teaching Assistant

Dec 2024 – Jun 2026

- Led lab sessions and mentored students across machine learning, robotics, and systems-oriented coursework.
- Guided students in designing experiments, debugging implementations, and interpreting results.
- Supported project-based coursework through one-on-one mentoring and group supervision.
- Contributed to evaluation design, including grading rubrics and assessment materials.

Precisely INC, Mumbai, IN

Software Support Engineer I: Data Integration

Jan 2022 – Aug 2024

- Supported deployments and upgrades across AWS, Azure, and GCP, resolving environment-specific runtime and compatibility issues.
- Led enterprise client onboarding by mapping data integration requirements to scalable distributed and cloud-based architectures.
- Supported proof-of-concept deployments and production transitions across ingestion, processing, and storage pipelines.
- Diagnosed production failures across Hadoop, HDFS, MS SQL Server, Oracle, and IBM i / AS400-based integration systems.
- Performed root cause analysis on distributed pipeline failures, improving system stability and reducing recurring operational issues.
- Managed cluster configurations and HDFS operations supporting large-scale distributed data processing workloads.
- Bridged customer requirements and technical implementation by guiding workflow design across distributed data pipelines.

SKILLS

Programming & Scripting: Python, Java, JavaScript (Node.js), SQL, Bash

Security, Systems & Infrastructure: gRPC, Docker, Distributed Systems, Cryptography (PyCryptodome), Oblivious Data Structures, REST APIs

Data Engineering & Databases: Hadoop, HDFS, Apache Kafka, Snowflake, Databricks, MS SQL Server, Oracle, IBM i (AS/400)

Cloud Platforms: AWS (EC2, S3, Lambda), Azure, GCP

Machine Learning & Data Science: PyTorch, TensorFlow, Scikit-learn, Pandas, NumPy, XGBoost, LightGBM, NLTK, OpenCV

LLM Systems & NLP: RAG Pipelines, Semantic Chunking, Vector Retrieval, Prompt Engineering, Multi-Model Evaluation, Embedding Pipelines

Responsible & Privacy-Aware AI: Explainable AI, Feature Attribution, SHAP-based Analysis, Fairness-Aware AI, Privacy-Preserving ML

Semantic Technologies: Knowledge Graphs, Ontologies, Semantic Web, Entity Linking, RDF/SPARQL Concepts, Wikidata, DBpedia

Research & Tools: Jupyter, LaTeX, Experimental Design, Statistical Analysis, Git, CI/CD, Docker Compose

RESEARCH PROJECTS

Selected project details, demos, and code links: [Available here](#)

OBFI: Implementation, Empirical Analysis, and Crossover Characterization of Competing Protocols

CWI Amsterdam | 2026

- Built a **distributed client-server system** in Python (gRPC, encryption, persistence) to implement and compare two cryptographic protocols.
- Translated formal theoretical protocol specifications into executable systems, resolving ambiguities in prior research.
- Designed and ran experiments across varying batch sizes and filter capacities to evaluate real-world system behavior for a sensitivity study.
- Developed a decision framework for selecting between protocols based on **deployment parameters and observed performance trends**.
- Ensured correctness and consistency through systematic testing across configurations.

Energy-Performance Trade-offs of Java Garbage Collection Strategies

VU Amsterdam | 2025

- Conducted a quantitative empirical study comparing Serial, Parallel, and G1 garbage collectors under controlled Java workloads using EnergiBridge and Intel RAPL.
- Designed an **RCBD-based experimental setup** with automated execution and data collection pipelines built in Python.
- Applied statistical methods (**ANOVA and Tukey HSD**) to evaluate energy efficiency and runtime trade-offs across configurations.
- Compared **OpenJDK and Oracle JDK** implementations to measure performance differences under varying workload conditions.
- Used quantitative analysis to identify system-level trade-offs between **throughput, latency, and energy consumption**.

Adaptive AI Interaction & Reasoning Platform

VU Amsterdam | 2026

- Built a full-stack RAG application supporting multi-model comparison across **LLaMA, Qwen, GPT, and Groq-hosted models**.
- Designed semantic chunking and vector retrieval pipelines with configurable **similarity metrics and ranked context inspection**.
- Developed **multi-round critique and evaluation pipelines** to analyse answer quality, reasoning consistency, and model behaviour.
- Implemented contextual highlighting, automated insight generation, side-by-side response comparison, and exportable evaluation logs.

PACSHAP-FeatureGraph: Privacy-Preserving Feature Selection Framework

UVA Amsterdam | 2024

- Developed a **privacy-preserving feature selection framework** aligned with **GDPR data minimization principles**.
- Combined **SHAP values** with **knowledge graphs** for interpretable and context-aware feature attribution.
- Implemented privacy-preserving mechanisms (**PAC-privacy, autoencoder-based encoding**) for secure data handling.
- Built an automated pipeline supporting numerical and categorical data for feature selection.
- Applied ML/NLP techniques (transformers, knowledge graph APIs) to evaluate feature importance and model behaviour.

Reactivity-Reflectivity Vector Framework

VU Amsterdam | 2025

- Designed a framework to study political polarization using qualitative concepts such as **framing, reflectiveness, and audience reactivity**.
- Collected and analyzed large-scale YouTube transcript and comment data across major political events and news outlets.
- Created structured comparisons across topics and media sources to **support fair and context-aware analysis**.
- Used NLP techniques to translate behavioural and engagement patterns into **measurable features**.
- Combined computational analysis with qualitative interpretation to study how media narratives and audience responses interact online.

ADDITIONAL EXPERIENCE & ACHIEVEMENTS

- Awards:** Winner AGBI Health-Tech Grand Challenge (NITI Aayog, 2022), Top-2 Finalist, TechBharat National Hackathon (2021); Best Employee Award, Spot Award, Joy Giver Award (Precisely, 2022-2024)
- Certifications:** Executive Post Graduate Certificate in Software Development with specialization Full Stack Development (IIIT Bangalore), AWS Solutions Architect Associate, Azure Data Engineer Associate, Snowflake SnowPro Core.
- Other:** Founder, ANR Medicus Technologies (2021–2022), HPAIR Delegate, French Language Olympiad Bronze Medalist
- Languages:** English (Fluent), Dutch (A2 working towards B1), French (A2).

REFERENCES

Dr. Marten van Dijk
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Wiskunde & Informatica)
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